THIS IS AN OPEN BOOK EXAMINATION
YOU MUST HAND-WRITE YOUR ANSWER

UNIVERSITY NO. _____

THE UNIVERSITY OF HONG KONG

DEPARTMENT OF CHEMISTRY

CHEM1043 General Chemistry II

May 30, 2020 (Saturday) **2:30 pm – 4:30 pm**

(Cutoff time for uploading your answer script to OLEX: 5:00 pm)

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

This paper consists of **six** questions (100 marks). Answer **ALL** questions.

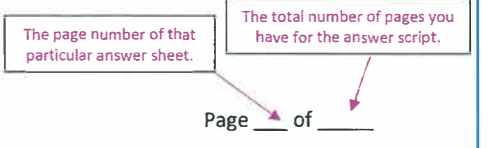
Hand-write your answers on A4 size white papers using black pen or blue pen.

Write on each answer sheet your student number and the page number.

A list of physical constants and a periodic table are provided on the last pages of this question paper.

At the end of the examination, please do the following:

- ☐ Check that you have written your **student number** and **page number** on each page of your answer script.



- ☐ Scan or take photos of your answer script and combine the pages into one single file.
- ☐ Check your file to make sure all pages are included and the images are clear.
- ☐ Submit your file through OLEX. And **make sure you have submitted the correct and complete file**.

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There are **six questions** in this paper. Answer **ALL** questions. **Hand-write** your answers on A4 paper.

Question 1 (16 marks)

(a) (i) List all the basic postulates of the kinetic-molecular theory of gases.

(ii) Further to (i), which assumption is most incorrect at very high pressures? Which one is most incorrect at very low temperatures? Explain your answers in each case.

(6 marks)

(b) (i) A 350.0-mL flask contains 0.1500 mol Ar at 24 °C. What is the pressure of the gas in kPa? Show your calculations.

(ii) Suppose 23.9 mg of BrF_3 (MW = 136.90 g/mol) exerts a pressure of 10.0 Torr at 100 °C. What is the volume of the container in mL? Show your calculations.

(iii) A 100.0-mL flask contains SO_2 (MW = 64.06 g/mol) at 0.77 atm and 30 °C. What mass of gas is present? Show your calculations.

(iv) A storage tank of volume 6.00×10^3 m^3 contains CH_4 at 129 kPa and 14 °C. How many moles of CH_4 are present? Show your calculations.

(10 marks)

Question 2 (16 marks)

(a) Consider a molecule of B_2 and that of F_2.

(i) Construct a well-labelled MO diagram for each of these two molecules. Hence, determine their bond orders and predict their magnetic properties.

(ii) Explain in detail any difference in their MO diagrams in terms of the **relative order of the molecular orbitals formed (i.e. the pattern of the MO diagrams)**.

(8 marks)

Question 2 (Con't)

(b) (i) In the context of band theory, what is a band? How does a band form?

(ii) Draw the band structure of potassium and that of silicon, respectively. Label the bands. With reference to these band structures, account for their electrical conductivity.

(8 marks)

Question 3 (21 marks)

(a) The density of a 2.45 M aqueous solution of CH_3OH is 0.976 g/mL. What is the molality of the solution? The molar mass of CH_3OH is 32.04 g. Show your calculations.

(4 marks)

(b) The solubility of nitrogen gas at 25 °C and 1 atm is 6.8×10^{-4} mol/L. What is the concentration, in molarity, of N_2 dissolved in water under atmospheric conditions? The partial pressure of N_2 gas in the atmosphere is 0.78 atm.

(2 marks)

(c) A solution was prepared by dissolving 20.0 g of a compound to 125 g of water at 25 °C. Assume this compound is a non-electrolyte and does not react with water. The vapour pressure of pure water is 23.76 Torr at 25 °C. The observed vapour pressure of the solution was found to be 22.67 Torr. Calculate the molar mass of the compound. Show your calculations.

(5 marks)

(d) Calculate the normal freezing point and boiling point of an antifreeze solution that is 50.0% by mass of ethylene glycol ($HOCH_2CH_2OH$, MW = 62.07 g/mol) in water. Given that for water, the freezing-point depression constant = $1.86°C \cdot mol^{-1} \cdot kg$, and the boiling point elevation constant = $0.52°C \cdot mol^{-1} \cdot kg$.

(5 marks)

(e) What is recrystallization commonly used for? Briefly discuss the working principle of recrystallization. (5 marks)

Question 4 (17 marks)

(a) Calculate the solubility, in molarity, of $Fe(OH)_3$ in an aqueous solution buffered at pH = 5.0. Assume the pH is constant and given that K_{sp} of $Fe(OH)_3 = 4 \times 10^{-38}$. Show the procedure by which you arrive at the answer.

(4 marks)

(b) (i) What are the criteria for a successful *fractional precipitation*? (3 marks)

(ii) A solution contains 1.0×10^{-4} M Cu^+ and 2.0×10^{-3} M Pb^{2+}. If a source of I^- is added to this solution gradually, will PbI_2 or CuI precipitate first? Explain your answer by specifying the concentration of I^- required to begin precipitation of each salt. Given that K_{sp} of $PbI_2 = 1.4 \times 10^{-8}$ and K_{sp} of CuI $= 5.3 \times 10^{-12}$.

(5 marks)

(c) In the presence of aqueous ammonia, copper(II) forms the complex ion $Cu(NH_3)_4^{2+}$. Determine the molar concentration of free (uncomplexed) copper(II) ion in solution when 0.10 mole of $Cu(NO_3)_2$ is dissolved in a litre of 3.0 M NH_3 (aq). Given that the formation constant (K_f) of $Cu(NH_3)_4^{2+} = 1.1 \times 10^{13}$.

(5 marks)

Question 5 (18 marks)

At 1000 K, an equilibrium mixture in the reaction $CO_2(g) + H_2(g) \rightleftharpoons CO(g) + H_2O(g)$ contains 0.203 mol CO_2, 0.290 mol H_2, 0.247 mol CO, and 0.157 mol H_2O.

(i) What is the equilibrium constant at 1000 K? Show your calculations. (3 marks)

(ii) Calculate the maximum non-expansion work at 1000 K. (3 marks)

(iii) In which direction would a spontaneous reaction occur if the following were brought together at 1000 K: 0.05 mol CO_2, 0.075 mol H_2, 0.1 mol CO, and 0.075 mol H_2O? Show your reasoning.

(6 marks)

(iv) In part (iii), find the number of mole of each species at equilibrium. Show your calculations.

(6 marks)

Question 6 (12 marks)

For the voltaic cell,

$Pt(s) \mid Cr^{3+}(aq), Cr^{2+}(aq) \parallel V^{3+}(aq), V^{2+}(aq) \mid C(s)$

given that $V^{3+}(aq) + e^- \rightleftharpoons V^{2+}(aq)$ $\quad E^\circ = -0.255$ V

$Cr^{3+}(aq) + e^- \rightleftharpoons Cr^{2+}(aq)$ $\quad E^\circ = -0.407$ V

determine the following:

[*show your calculations for* (ii) – (iv)]

(i) the equation for the cell reaction (2 marks)

(ii) E^0_{cell} (2 marks)

(iii) $\Delta_r G^\circ$ (3 marks)

(iv) K (3 marks)

(v) Indicate whether the reaction goes essentially to completion or to a limited extent only in their standard states. (2 marks)

– END OF PAPER –

Physical Constants

Planck's constant	$h = 6.626 \times 10^{-34}$ J·s·molecule^{-1}
Velocity of light in vacuum	$c = 2.998 \times 10^{8}$ m·s^{-1}
Avogadro's number	$N = 6.022 \times 10^{23}$ molecules·mol^{-1}
Boltzmann constant	$k_B = 1.381 \times 10^{-23}$ J·K^{-1}·molecule^{-1}
Gas constant	$R = 8.314$ J·mol^{-1}·K^{-1}
	$= 0.08206$ atm·dm^3·mol^{-1}·K^{-1}
	$= 0.08314$ bar·dm^3·mol^{-1}·K^{-1}
	$= 8.314$ Pa·m^3·mol^{-1}·K^{-1}
Faraday constant	$F = 96487$ C·mol^{-1} (of electrons transferred)
	$= 23061$ cal·volt^{-1}·mol^{-1} (of electrons transferred)
Elementary charge	$e = 1.602 \times 10^{-19}$ C
Mass of proton	$m_p = 1.673 \times 10^{-27}$ kg
Mass of electron	$m_e = 9.109 \times 10^{-31}$ kg
Standard gravitational acceleration	$g = 9.807$ m·s^{-2}
Ion product of water	$K_w = 1.0 \times 10^{-14}$ (at 25 °C)

Some Equations for Kinetic Theory of Gases:		
$u_{rms} = \sqrt{\frac{3RT}{M}}$	$\overline{E_k} = \frac{3}{2}\left(\frac{RT}{N_A}\right)$	$P = \frac{1}{3}\frac{N}{V}\overline{mu^2}$

Conversion Table:	**Numerical Constants:**
Specific heat capacity of water = 1 cal	$\pi = 3.141593$
= 4.184 J	
1 erg = 10^{-7} J	e = 2.718282
1 G = 10^{-4} T (G = gauss, T = tesla)	ln x = 2.303 log x
1 atm = 101,325 Pa = 760 mmHg (Torr)	
1 bar = 10^5 Pa	
0 °C = 273.15 K	

The periodic table

1	2		3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Hydrogen 1 **H** 1.008																		Helium 2 **He** 4.0026
Lithium 3 **Li** 6.94	Beryllium 4 **Be** 9.0122			Key: Element Name Atomic number **Symbol** Atomic weight (mean relative mass)									Boron 5 **B** 10.81	Carbon 6 **C** 12.011	Nitrogen 7 **N** 14.007	Oxygen 8 **O** 15.999	Fluorine 9 **F** 18.998	Neon 10 **Ne** 20.180
Sodium 11 **Na** 22.990	Magnesium 12 **Mg** 24.305												Aluminium 13 **Al** 26.982	Silicon 14 **Si** 28.085	Phosphorus 15 **P** 30.974	Sulfur 16 **S** 32.06	Chlorine 17 **Cl** 35.45	Argon 18 **Ar** 39.948
Potassium 19 **K** 39.098	Calcium 20 **Ca** 40.078(4)		Scandium 21 **Sc** 44.956	Titanium 22 **Ti** 47.867	Vanadium 23 **V** 50.942	Chromium 24 **Cr** 51.996	Manganese 25 **Mn** 54.938	Iron 26 **Fe** 55.845(2)	Cobalt 27 **Co** 58.933	Nickel 28 **Ni** 58.693	Copper 29 **Cu** 63.546(3)	Zinc 30 **Zn** 65.38(2)	Gallium 31 **Ga** 69.723	Germanium 32 **Ge** 72.63	Arsenic 33 **As** 74.922	Selenium 34 **Se** 78.96(3)	Bromine 35 **Br** 79.904	Krypton 36 **Kr** 83.798(2)
Rubidium 37 **Rb** 85.468	Strontium 38 **Sr** 87.62		Yttrium 39 **Y** 88.906	Zirconium 40 **Zr** 91.224(2)	Niobium 41 **Nb** 92.906(2)	Molybdenum 42 **Mo** 95.96(2)	Technetium 43 **Tc** [97.91]	Ruthenium 44 **Ru** 101.07(2)	Rhodium 45 **Rh** 102.91	Palladium 46 **Pd** 106.42	Silver 47 **Ag** 107.87	Cadmium 48 **Cd** 112.41	Indium 49 **In** 114.82	Tin 50 **Sn** 118.71	Antimony 51 **Sb** 121.76	Tellurium 52 **Te** 127.60(3)	Iodine 53 **I** 126.90	Xenon 54 **Xe** 131.29
Caesium 55 **Cs** 132.91	Barium 56 **Ba** 137.33	57-70 *	Lutetium 71 **Lu** 174.97	Hafnium 72 **Hf** 178.49(2)	Tantalum 73 **Ta** 180.95	Tungsten 74 **W** 183.84	Rhenium 75 **Re** 186.21	Osmium 76 **Os** 190.23(2)	Iridium 77 **Ir** 192.22	Platinum 78 **Pt** 195.08	Gold 79 **Au** 196.97	Mercury 80 **Hg** 200.59	Thallium 81 **Tl** 204.38	Lead 82 **Pb** 207.2	Bismuth 83 **Bi** 208.98	Polonium 84 **Po** [208.98]	Astatine 85 **At** [209.99]	Radon 86 **Rn** [222.02]
Francium 87 **Fr** [223.02]	Radium 88 **Ra** [226.03]	89-102 **	Lawrencium 103 **Lr** [262.11]	Rutherfordium 104 **Rf** [265.12]	Dubnium 105 **Db** [268.13]	Seaborgium 106 **Sg** [271.13]	Bohrium 107 **Bh** [270]	Hassium 108 **Hs** [277.15]	Meitnerium 109 **Mt** [276.15]	Darmstadtium 110 **Ds** [281.16]	Roentgenium 111 **Rg** [280.16]	Copernicium 112 **Cn** [285.17]	Ununtrium 113 **Uut** [284.18]	Flerovium 114 **Fl** [289.19]	Ununpentium 115 **Uup** [288.19]	Livermorium 116 **Lv** [293]	Ununseptium 117 **Uus** [294]	Ununoctium 118 **Uuo** [294]

*lanthanoids	Lanthanum 57 **La** 138.91	Cerium 58 **Ce** 140.12	Praseodymium 59 **Pr** 140.91	Neodymium 60 **Nd** 144.24	Promethium 61 **Pm** [144.91]	Samarium 62 **Sm** 150.36(2)	Europium 63 **Eu** 151.96	Gadolinium 64 **Gd** 157.25(3)	Terbium 65 **Tb** 158.93	Dysprosium 66 **Dy** 162.50	Holmium 67 **Ho** 164.93	Erbium 68 **Er** 167.26	Thulium 69 **Tm** 168.93	Ytterbium 70 **Yb** 173.05
**actinoids	Actinium 89 **Ac** [227.03]	Thorium 90 **Th** 232.04	Protactinium 91 **Pa** 231.04	Uranium 92 **U** 238.03	Neptunium 93 **Np** [237.05]	Plutonium 94 **Pu** [244.06]	Americium 95 **Am** [243.06]	Curium 96 **Cm** [247.07]	Berkelium 97 **Bk** [247.07]	Californium 98 **Cf** [251.08]	Einsteinium 99 **Es** [252.08]	Fermium 100 **Fm** [257.10]	Mendelevium 101 **Md** [258.10]	Nobelium 102 **No** [259.10]